

Answer Key & Teacher Notes

Session 1 – Polygons and Perimeter on the Plane • Session 2 – Designing Polygons with Scale

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

Polygons and Perimeter on the Plane

TALK PROMPTS

- How do matching coordinates tell you which sides run across and which run up and down?
- Why does finding only two side lengths give you the whole perimeter of a rectangle?

PARTNER TASK

Garden Plot (Form A, problem 2). The plot has corners $(-4, 6)$, $(9, 6)$, $(9, -3)$ and $(-4, -3)$. One partner finds the top side; the other finds the left side. Together find the perimeter ($13 + 9 + 13 + 9 = 44$ m) and explain why two side lengths give the whole perimeter.

WATCH FOR

- Subtracting coordinates with opposite signs ($(-5, 4)$ to $(3, 4)$ as 2 instead of 8).
- Adding only two sides of a rectangle for the perimeter.
- Mixing up perimeter (units) and area (square units), or forgetting the $1/2$ in triangle area.

SESSION 2 • TEACHER NOTES

Designing Polygons with Scale

TALK PROMPTS

- How does the scale change your answer even though the drawing stays the same size?
- Why should you change each side to real length before you find the area?

PARTNER TASK

Fence Cost (Form A, problem 6). The yard has corners $(-4, 3)$, $(8, 3)$, $(8, -5)$ and $(-4, -5)$; each unit is 4 feet. One partner finds the side lengths in grid units; the other changes them to feet. Together find the perimeter (160 ft) and check it by multiplying the grid perimeter of 40 units by 4.

WATCH FOR

- Leaving a side in grid units (12 units written as 12 ft).
- Multiplying the grid area by the scale only once instead of changing both sides to real length.
- Multiplying instead of dividing to find a missing side from an area.

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.G.A.3	1	2	1	2	2	2	2	3	3	1	1
S1 • Form B	6.G.A.3	1	2	1	2	2	2	2	3	3	1	1
S2 • Form A	6.G.A.3	1	2	2	1	1	2	2	3	3	1	1
S2 • Form B	6.G.A.3	1	2	2	1	1	2	2	3	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

polygon: A closed shape with **straight** sides.

vertex: A **corner** point of a polygon.

perimeter: The distance all the way **around** a shape.

area: The number of **square** units inside a shape.

- Two vertices that share a coordinate make a **horizontal** or vertical side.
- Perimeter is the **sum** of all the sides. Area of a rectangle is length $\times$ width.

Example: Top side: same y , opposite signs, so $1 + 5 = 6$ units. Right side: same x , opposite signs, so $4 + 3 = 7$ units. Perimeter: $6 + 7 + 6 + 7 = 26$ units. Answer: Area = $6 \times 7 = 42$ square units

YOU TRY

- Each side is $3 + 1 = 4$ units; $P = 4 \times 4 = 16$ units
- $(-4, -5)$

EXIT TICKET

- a. $6 + 3 = 9$ units b. $2 + 4 = 6$ units
- $P = 9 + 6 + 9 + 6 = 30$ units; $A = 9 \times 6 = 54$ square units
- A** (adding all the sides)

PRACTICE FORM A

- a. $3 + 6 = 9$ units b. $4 + 2 = 6$ units c. $9 + 6 + 9 + 6 = 30$ units d. $9 \times 6 = 54$ square units
- a. rectangle plotted b. $4 + 9 = 13$ m c. $6 + 3 = 9$ m d. $13 + 9 + 13 + 9 = 44$ m
- $7 + 11 = 18$ m and $10 + 1 = 11$ m; $P = 18 + 11 + 18 + 11 = 58$ m
- $(-2, 1)$
- base $5 + 7 = 12$, height $2 + 5 = 7$; $1/2 \times 12 \times 7 = 42$ square units
- $6 + 7 = 13$ yd and $9 - 2 = 7$ yd; $13 \times 7 = 91$ square yards
- $12 + 6 + 5 + 6 + 7 + 12 = 48$ ft
- a. 15 units by 9 units b. 48 units; $48 \times 7 = \$336$ c. 135 square units; $135 \times 2 = \$270$
- The x -coordinates -5 and 3 have opposite signs, so the side crosses the y -axis. Jorge should add: $5 + 3 = 8$ units.
- B** (26 units)
- C** (40 square units)

Self-check word VERTEX

PRACTICE FORM B

- a. $5 + 5 = 10$ units b. $3 + 4 = 7$ units c. $10 + 7 + 10 + 7 = 34$ units d. $10 \times 7 = 70$ square units
- a. rectangle plotted b. $5 + 7 = 12$ m c. $5 + 4 = 9$ m d. $12 + 9 + 12 + 9 = 42$ m
- $6 + 10 = 16$ m and $9 + 3 = 12$ m; $P = 16 + 12 + 16 + 12 = 56$ m
- $(-3, 2)$
- base $4 + 6 = 10$, height $3 + 5 = 8$; $1/2 \times 10 \times 8 = 40$ square units
- $5 + 8 = 13$ yd and $7 - 1 = 6$ yd; $13 \times 6 = 78$ square yards
- $11 + 6 + 4 + 6 + 7 + 12 = 46$ ft
- a. 14 units by 9 units b. 46 units; $46 \times 6 = \$276$ c. 126 square units; $126 \times 3 = \$378$
- The x -coordinates -9 and 6 have opposite signs, so the side crosses the y -axis. Bri should add: $9 + 6 = 15$ units.
- B** (26 units)
- C** (36 square units)

Self-check word CORNER

GUIDED NOTES

scale: How much real length one grid **unit** stands for.

area: The flat space **inside** a shape, in square units.

dimensions: The **length** and width of a shape.

- Multiply grid units by the **scale** to get real length.
- To find a missing side, **divide** the area by the known side.

Example: Real width: $5 \times 4 = 20$ m. Real height: $3 \times 4 = 12$ m. Answer: Area = $20 \times 12 = 240$ m²

YOU TRY

1. a. $8 \times 5 = 40$ ft b. $35 \div 5 = 7$ units
2. $600 \div 20 = 30$ ft

EXIT TICKET

1. a. 27 m b. 16 units
2. $20 \text{ ft} \times 30 \text{ ft} = 600 \text{ ft}^2$
3. **B** (divide)

PRACTICE FORM A

1. a. 24 ft b. 54 ft c. 7 units d. 15 units
2. a. $(-5, -3)$ b. 11 units = 110 ft; 9 units = 90 ft c. $110 \times 90 = 9,900 \text{ ft}^2$
3. 12 units = 36 ft; 9 units = 27 ft; $36 \times 27 = 972 \text{ ft}^2$
4. 150 ft and 100 ft; $150 + 100 + 150 + 100 = 500$ ft
5. $30,000 \div 150 = 200$ ft
6. 12 units = 48 ft; 8 units = 32 ft; P = 160 ft; $160 \times 6 = \$960$
7. 12 units = 24 ft; $384 \div 24 = 16$ ft
8. a. $8 \times 15 = 120$ ft b. $5,400 \div 120 = 45$ ft c. $45 \div 15 = 3$ units d. $120 + 45 + 120 + 45 = 330$ ft
9. 12 is the length in grid units $(3 + 9)$. Talia forgot the scale. Multiply: $12 \times 6 = 72$ feet.
10. **A** (30 ft)
11. **C** (100 feet)

Self-check word DESIGN

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 • POLYGONS AND PERIMETER ON THE PLANE

1. A $(-4, 3)$, B $(2, 3)$, C $(2, -2)$, D $(-4, -2)$. a. $4 + 2 = 6$ units b. $3 + 2 = 5$ units c. $6 + 5 + 6 + 5 = 22$ units d. $6 \times 5 = 30$ square units
2. a. $3 + 4 = 7$ units b. $2 + 4 = 6$ units c. $\frac{1}{2} \times 7 \times 6 = 21$ square units
3. Sides $5 + 6 = 11$ m and $3 + 4 = 7$ m; P = $11 + 7 + 11 + 7 = 36$ m
4. Each side is $3 + 2 = 5$ units; P = $4 \times 5 = 20$ units
5. $(-2, -1)$. It must line up with $(-2, 5)$ vertically ($x = -2$) and with $(3, -1)$ horizontally ($y = -1$).

PRACTICE FORM B

1. a. 40 ft b. 56 ft c. 6 units d. 12 units
2. a. $(-6, -4)$ b. 12 units = 120 ft; 9 units = 90 ft c. $120 \times 90 = 10,800 \text{ ft}^2$
3. 12 units = 24 ft; 9 units = 18 ft; $24 \times 18 = 432 \text{ ft}^2$
4. 100 ft and 60 ft; $100 + 60 + 100 + 60 = 320$ ft
5. $4,200 \div 70 = 60$ ft
6. 12 units = 36 ft; 7 units = 21 ft; P = 114 ft; $114 \times 8 = \$912$
7. 9 units = 18 ft; $360 \div 18 = 20$ ft
8. a. $6 \times 10 = 60$ ft b. $4,800 \div 60 = 80$ ft c. $80 \div 10 = 8$ units d. $60 + 80 + 60 + 80 = 280$ ft
9. 12 is the length in grid units $(7 + 5)$. Kenji forgot the scale. Multiply: $12 \times 8 = 96$ feet.
10. **B** (30 m)
11. **C** (90 meters)

Self-check word LAYOUT

SHEET 2 • DESIGNING POLYGONS WITH SCALE

1. A $(-3, 4)$, B $(-3, -2)$. a. $4 + 2 = 6$ units b. $6 \times 5 = 30$ ft c. $1,200 \div 30 = 40$ ft d. $40 \div 5 = 8$ units
2. a. 28 m b. 9 units
3. $14 \text{ ft} \times 10 \text{ ft} = 140 \text{ ft}^2$
4. Grid sides $2 + 2 = 4$ units (12 ft) and $4 + 1 = 5$ units (15 ft). P = $12 + 15 + 12 + 15 = 54$ ft
5. Lily subtracted $5 - 4$. The signs differ, so add: $4 + 5 = 9$ units; $9 \times 3 = 27$ ft.
6. **B** (30 m)
7. **A** (20 ft)